

The Development Application Base of Engineering Talents (iDABET)
Project of the Industrial Development Administration, Ministry of Economic Affairs
**2025 Practical Ability Development Implementation Unit Project –
Engineering Talent Practical Ability Development Record Form**

Project No. of the Practical Ability Development Implementation Unit: DABET11403

No. of Engineering Talent: 40

Name of the Practical Ability Development Implementation Unit: 財團法人金屬工業研究發展中心/光電技術組

Project Name of the Practical Ability Development Implementation Unit: 半導體產業之真空設備與製程能力發展與人工智慧控制應用

Name of the Engineering Talent: 蘇翼

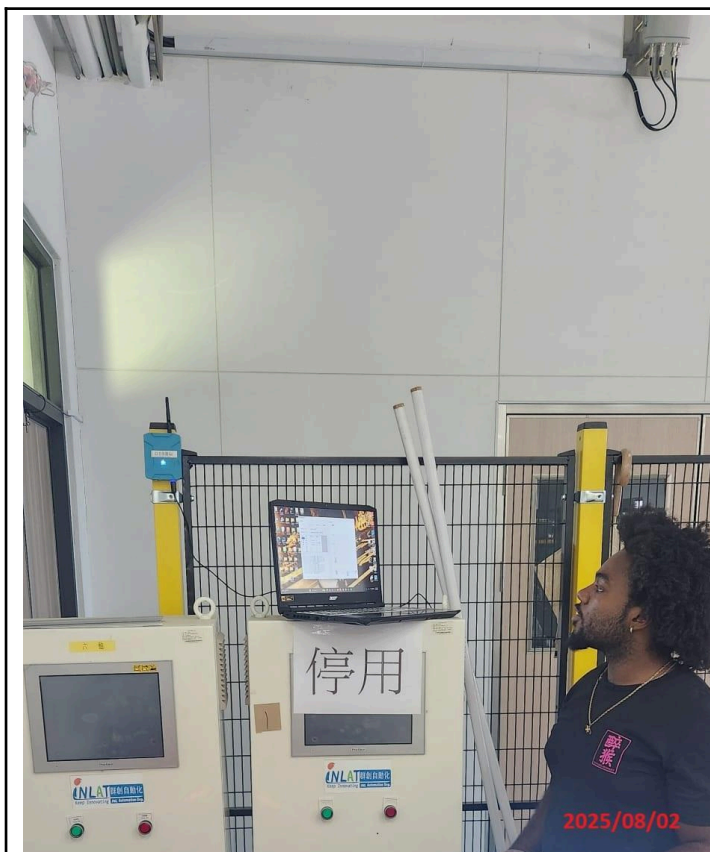
Item	Date	Instructor	No. of Hours (Note 1)	Type of Practical Ability Development Topic (Note 2)	Topic	Description
1	8/2	李宗信	5	1	Indoor Positioning System Research (Thesis)	Independent Study: Taking more test within the better perimeter to analyze the error.
2	8/5	李宗信	4	1	Final Report Presentation	Presenting my experience and progress throughout the 6 months of the internship.
3	8/6	李宗信	4	2	CVD Practice	Chemical Vapor Deposition practice at MIRDC
4	8/7	李宗信	2	1	3D Printing and Laser Engraving Technology(Theory)	Introduction to 3D printing and laser engraving.
5	8/7	李宗信	2	2	3D Printing and Laser Engraving Technology(Practical)	Application of 3D printing and laser engraving.
6	8/8	李宗信	4	1	Indoor Positioning System Research (Thesis)	Independent Study: Syncing the tag positioning system with the hexapod.
7	8/9	李宗信	3	1	Indoor Positioning System Research (Thesis)	Independent Study: Syncing the tag positioning system with the hexapod.
Total Hours on Practical Ability Development Topics			Total <u>20</u> hours.			

Please add rows to the table as needed.

Notes: (1) The hours for implementing practical ability development topics should be filled in according to the actual implementation status between March 1 and August 31.

- (2) Type of practical ability development topics: 1. Practice-based; 2. Industry collaboration-based.
- (3) Please attach at least 6 dated photos (yyyy/mm/dd) as evidence of the engineering talent's participation in practical ability development topics or activities.
- (4) Both the instructor and engineering talent shall sign at the bottom of the record form.
- (5) To implement the “2025 Development Application Base of Engineering Talents (iDABET) Project”, the Industrial Development Administration, Ministry of Economic Affairs will collect, process and use the personal data of the party involved without notifying the said party, as prescribed in Subparagraph 2 of Paragraph 2 of Article 8 of the Personal Data Protection Act.

【Photos of Participation in Practical Ability Development Topics or Activities】



I returned to the initial testing area to reanalyze the perimeter that yielded the lowest positioning error during previous trials. By closely examining the UWB protocol's performance in this setup, I aim to identify the contributing factors and implement control measures to further minimize error in future indoor positioning experiments.



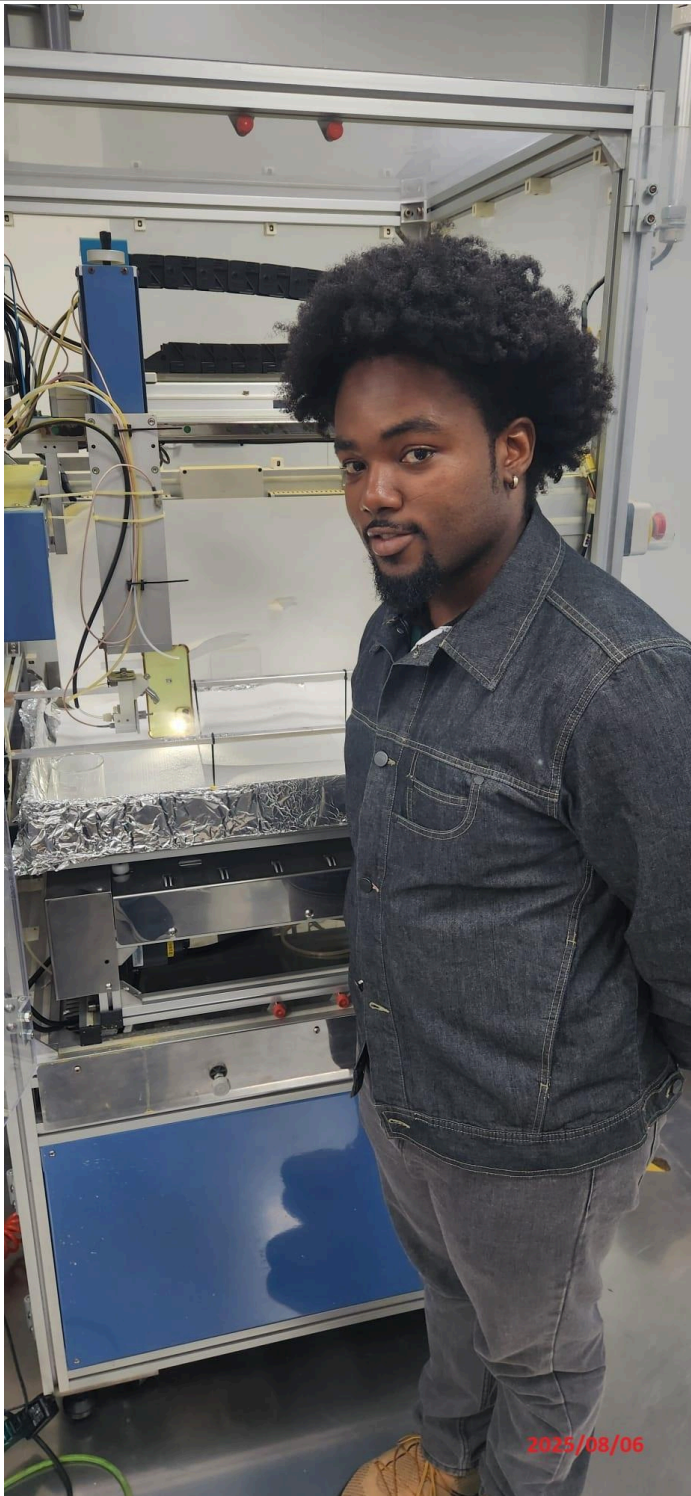
Final Report Presentation

Today was the final report presentation. I felt a little nervous at the start; however, just moments before I began, the nerves disappeared and the information flowed smoothly. I received great feedback from the other professors, which was both encouraging and motivating.



Presentation Award

After the presentation, I was pleasantly surprised to receive the 2nd place award for outstanding performance. I am truly grateful for this opportunity to learn and grow as an engineer. This experience has provided me with valuable insights into the semiconductor industry and deepened my understanding of the step-by-step process involved in manufacturing an IC chip.



CVD Extra Practice

Today, I returned to the CVD (Chemical Vapor Deposition) department to deepen my understanding of the process. CVD is a critical step in semiconductor manufacturing, where thin films are deposited onto a substrate through chemical reactions of gaseous precursors. This technique allows for precise control over film thickness, uniformity, and composition, which is essential for creating high-performance IC chips. During my visit, I observed how factors such as temperature, gas flow rate, and chamber pressure influence the quality of the deposited layers. I also learned about different CVD methods, including LPCVD (Low-Pressure CVD) and PECVD (Plasma-Enhanced CVD), and how they are selected based on the specific material and application requirements. This hands-on exposure gave me a clearer picture of the role CVD plays in ensuring the reliability and efficiency of semiconductor devices.



3D Printing Introduction and Application

Here, we explored the 3D printing process in detail, starting with the essential pre-print preparations required in the software. This included setting the correct print parameters such as material type, layer height, print speed, and support structures to ensure a successful outcome. I actively participated in the session, answering most of the questions correctly, and was honored to receive a grand prize in recognition of my effort and engagement. This experience not only enhanced my technical knowledge but also strengthened my confidence in applying 3D printing skills effectively.



Laser Engraving Introduction and Application

We learned about laser engraving and cutting techniques, along with their wide range of practical applications. Although I was not very familiar with this topic beforehand, I found the session highly engaging and insightful. The precision, versatility, and creative possibilities of laser technology sparked my interest, and I am eager to further explore its capabilities in future projects.

Notes:

1. Please add rows to the table as needed. Each photo shall be dated at the bottom right, and each photo shall be accompanied by descriptions below it to explain the topic.
2. Please provide the original files of 1 to 3 photos (JPEG files, each above 1MB in size and with a resolution above 300 DPI).